

Palla Pavan Kumar Reddy

AI & Full Stack Developer | Python / Flask / IoT

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SUMMARY

AI student who ships full-stack, not just models. Built and deployed a live voice-enabled AI assistant (Python/Flask/Web Speech API) and a remote voice-controlled IoT relay system on ESP32. Comfortable owning a project end-to-end — backend, frontend, and the hardware layer when a project needs it.

TECHNICAL SKILLS

Languages: Python, Java

Frontend: HTML5, CSS3, JavaScript

Backend: Flask, REST APIs

AI / ML: Generative AI, Prompt Engineering, Speech Recognition, Conversational AI

IoT / Embedded: ESP32, ESP8266, C++ (Arduino), Embedded Systems, Sensor Integration

Practices: Responsive Design, Debugging Across Hardware/Software Boundaries, Problem Solving

PROJECTS

AI Assistant Platform — Completed & Deployed

Python, Flask, JavaScript, Web Speech API, HTML5/CSS3 | ai-assitent-lijk.onrender.com

- Built and deployed a full-stack conversational assistant supporting typed and spoken input, live on Render.
- Implemented real-time speech-to-text/text-to-speech with cross-browser fallback logic for Chrome and Edge.

Voice-Controlled IoT Device Management System — Completed

Python, Flask, ESP32, C++ (Arduino), REST APIs, Voice Recognition

- Built a Flask relay server enabling ESP32 devices to be voice-controlled over the internet, not just locally.
- Designed a heartbeat/reconnect protocol so devices recover automatically from dropped WiFi connections.

AI YouTube Video Automation Platform — In Testing

Python, AI Script/Voice/Image Generation, Automation Dashboard | github.com/pavankumarreddy2006/youtube-

- Architected a 5-stage content pipeline (script, voice, visuals, subtitles, render) with independently swappable stages.
- Stabilized script and narration stages; added a manual review checkpoint to stop errors compounding downstream.

Smart Punching Bag — Prototype

Embedded Systems, Sensors, Microcontroller Firmware, AI-Assisted Analytics

- Built a bench-test sensor rig measuring punch force, speed, count and reaction time in real time.
- Currently solving sensor-noise filtering to separate bag swing-back from genuine repeat hits.

EDUCATION

B.Sc., Artificial Intelligence — Expected 2027

Ideal College of Arts and Sciences, Adikavi Nannaya University | Aggregate: 70%

Intermediate (MPC)

Aditya Junior College | Aggregate: 60%

SSC

Pragati English Medium School | Aggregate: 75%

CERTIFICATIONS

- Generative AI for Software Development — IBM SkillsBuild
- Python Programming — Infosys Springboard
- Technology Certification — EY

Languages: English, Telugu